

Unit 05 Practice Problems: Continuous Random Variables

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Required Practice Problems

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and/or preparing for exams, make sure you have completed and reviewed all practice problems from the unit. The **solutions** are posted on Canvas ([Unit-PP-05-06-Solution.pdf](#) > Practice Problem-Solution > STAT 312 > Files).

5.1

A continuous random variable X has the following pdf:

$$f(x) = \begin{cases} -\frac{2}{9}x + \frac{2}{3}, & 0 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Draw the pdf. Confirm the pdf integrates to 1.

(b) Find $P(1 < X < 3)$.

5.2

A continuous random variable X has the following pdf:

$$f(x) = \begin{cases} -2x + 4, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Draw the pdf. Confirm the pdf integrates to 1.

(b) Find $P(1 < X < 1.4)$.

5.3

A continuous random variable X has the following pdf:

$$f(x) = \begin{cases} \frac{1}{4}x^3, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Draw the pdf. Confirm the pdf integrates to 1.

(b) Find $P(0.9 < X < 1.2)$. Calculus required.

5.4

Refer to Practice Problem 5.1. Suppose a data set is made up of $n = 1000$ independent observations of X .

- (a) If $\bar{X}$, s^2 , and s were calculated for the data set, they will be approximately equal to what values?
- (b) Approximately how many of the 1000 observations would you expect to be greater than 1?

5.5

Refer to Practice Problem 5.2. Find μ , σ^2 , and σ .

5.6

Refer to Practice Problem 5.3. Find μ , σ^2 , and σ .

5.7

X = circumference of a randomly selected baseball (inches)

$$X \sim \text{UNIF}(9.0, 9.1)$$

- (a) What is the probability a randomly selected baseball has a circumference greater than 9.035 inches?
- (b) Find the population mean, variance, and standard deviation of X .

5.8

An artist has a large collection of colored pencils. If one colored pencil is selected at random, its length is a Uniformly distributed random variable on the interval 6 inches to 7.2 inches. What is the probability a randomly selected colored pencil is between 6.3 and 6.7 inches long?

5.9

Emma is watching a severe thunderstorm. X = the waiting time (seconds) until the next lightning strike. X is Exponentially distributed with a mean waiting time of 5 seconds.

- (a) Since X is Exponentially distributed with a mean waiting time of 5 seconds, what is the value of λ ?
- (b) What is the variance and standard deviation among the waiting times?
- (c) What is the probability Emma will observe the next lightning strike within the next 7 seconds?
- (d) What is the probability Emma will not observe any lightning strikes within the next 10 seconds?

5.10

A radioactive substance is rapidly emitting particles. X_{POI} = the # of particles emitted per second. X_{POI} is a Poisson random variable with a mean of $\lambda = 2.5$ particles per second. Therefore, the waiting time until the next particle emission is Exponentially distributed. X_{EXP} is Exponential with a mean waiting time of

$$\frac{1}{\lambda} = \frac{1}{2.5} = 0.4 \text{ seconds.}$$

- (a) What is the probability we will wait at least one second until the next particle is emitted?
- (b) What is the probability the next particle will be emitted sometime between 0.28 and 0.35 seconds from now?
- (c) Given there are no particle emissions in the next 0.20 seconds, what is the conditional probability there will be no particle emissions in the next 0.26 seconds?
- (d) (Optional) Find the probability there will be no particle emissions for the next 0.06 seconds and compare it to your answer to the previous question. The Exponential random variable has a Memoryless Property:

$$P(X_{\text{EXP}} > t + t_0 \mid X_{\text{EXP}} > t_0) = P(X_{\text{EXP}} > t).$$

- (e) (Optional) Do you drive a new car or an old car? Now that you know about the Memoryless Property of the Exponential random variable, do you believe the lifetimes of cars are Exponentially distributed? Why or why not? Discuss.

5.11

Solve the following exercises from the “Chapter 4 Exercises” section of the textbook (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger).

- 4.1.1, 4.1.4, 4.1.5, 4.1.9, 4.2.5, and 4.2.8.

[To find these exercises in the digital copy of the textbook (**Textbooks > STAT 312 > Files > Canvas**), go to the xvi Contents page and click the “EXERCISES” tab located at the lower-right corner of the page. This will open the complete list of chapter exercises. The **solutions** are posted on Canvas (**textbook-solution.pdf > Practice Problem-Solution > STAT 312 > Files**).]

Supplemental Practice Problems

The problems below provide additional practice opportunities for students who would like more practice with the concepts covered in this unit.

Chapter 2 (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger)

- 4.2.1, 4.2.9, 4.2.11, 4.3.1, 4.3.2, 4.3.6, 4.4.1, 4.4.3, and 4.4.5.